Gour Gupal Talukder Shawon

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Portfolio | LinkedIn | github.com/gfor-gour | Google Scholar

Education

B.Sc. in Software Engineering *Mar 2022 – Sep 2026*
Shahjalal University of Science and Technology Sylhet, Bangladesh

- **CGPA: 3.52** / 4.00 | Final 3 Semesters: **3.78** / 4.00
- **Thesis:** *Commit Message Generation using LLMs with Issue-Based Context* (Grade: 4.00 / 4.00)

Publications

1. **AgentCollabBench: Diagnosing When Good Agents Make Bad Collaborators**
   Aritra Mazumder, . . . , Gour Gupal Talukder Shawon, . . . (8th of 13 authors)
   ***NeurIPS** 2026, Evaluations & Datasets Track* (accepted); *FAGEN Workshop, ICML 2026* (accepted)
   arXiv | Dataset
2. **Commit Message Generation using LLMs with Issue-Based Context**
   Amit Kumar Sharma*, Gour Gupal Talukder Shawon*†, Prottya Roy Chayan, Mahfuzur Rahman Emon
   Informatica (under review); *co-first authors, †corresponding author
   Dataset

Research Experience

HiveMind Research Group *Remote*
Research Contributor Jan 2026 - Present

- **AgentCollabBench** (NeurIPS 2026): Worked on dataset design, task construction and evaluation setup for a 900-task benchmark that shows where multi-agent LLM systems silently drop constraints, spread false beliefs or leak private context.
- Found that communication topology accounted for 7–40% of the variance in constraint preservation during multi-hop forwarding.
- **Memory-gated tool use** (manuscript in preparation): Annotated and quality-checked coding-domain examples for a 340-pair benchmark examining when agents should apply stored rules or seek human confirmation before tool use (five annotators per pair; coding $\kappa = 0.72$).

Undergraduate Thesis *May 2025 – Nov 2025*
Commit Message Generation using LLMs with Issue-Based Context Co-first & corresponding author
Supervisor: Mahfuzur Rahman Emon, Lecturer, Software Engineering, IICT, SUST (on study leave)

- Designed the study, wrote the manuscript and curated 309 issue-linked commits from open-source repositories and compared GPT-4 Turbo, Claude 3.5 and Gemini 1.5 Pro.
- Evaluated outputs with automated text metrics and double-blind ratings from 30 software developers.

Industry Experience

RunAgent Technology Limited *Dhaka, Bangladesh*
Full Stack Engineer & FDE (Hybrid) Oct 2026 - Present

- Built and deployed a client-facing business system using **Next.js**, **TypeScript** and **PostgreSQL**, taking responsibility for implementation from initial development through production delivery.
- Manage staging and production deployments on a self-hosted **Coolify** server, maintaining a live system for client use and feedback.
- Presented the first working version to the client and am extending the system to a companion Expo (**React Native**) app that shares the existing backend and database.

ShellBeeHaken Ltd. *Dhaka, Bangladesh*
Intern Software Engineer (Onsite) Dec 2025 - May 2026

- Contributed to **production UI components** and navigation flows using **React Native (Expo)** and **Next.js**, integrating **REST APIs** and managing application state across screens.
- Assisted in building **event-driven backend services** with **AWS Lambda**; gained hands-on exposure to distributed systems concepts including **data replication** and **load balancing** during architecture discussions.
- Participated in the full **API development cycle** from writing data contracts to **integration testing**, while working in an **Agile** team with sprint planning and code reviews.

Competitive Programming

- Solved **1100+** programming problems across LeetCode, Vjudge, and Codeforces with a focus on algorithms, data structures, and problem-solving.
- Maintained a **433**-day LeetCode streak, earned **23** badges, and achieved a peak rating of **1622**.

Projects

Personality Detection Using Machine Learning GitHub | Live

- Improved model **F1 score by 20%** via advanced **feature engineering** and **hyperparameter tuning**, achieving **91% accuracy** in real-time personality prediction.
- Tools: Python, scikit-learn, Flask, Next.js, Pandas, NumPy, joblib.

Local Household Service Provider Platform GitHub | Live

- Designed an **optimized data-fetching architecture** that reduced retrieval time by **60%**, resulting in **20% higher** overall application responsiveness and user satisfaction.
- Integrated **Google Gemini-powered AI chatbot**, improving service query response rate by **60%**; authored critical system documents including SRS and project proposal.
- Tools: Next.js, Express.js, MongoDB, Nodemailer, Cloudinary.

Real-Time Multi-Module Chat Application GitHub

- Built a **real-time desktop chat system** using **Java, Socket.IO (Netty), PostgreSQL, and Swing**, structured with a **multi-module Maven architecture** and shared DTOs for consistent client-server communication.
- Implemented **WebSocket-based messaging** with a custom **reconnection strategy (exponential backoff)** and heartbeat monitoring to ensure resilient real-time connectivity.
- Designed an **event-driven backend architecture** supporting authentication, messaging, and file transfer workflows, with secure and efficient persistence using **HikariCP, PostgreSQL indexing, and BCrypt authentication**.

Research & Technical Skills

Languages	Python, C++, TypeScript, JavaScript, Java
ML & NLP	PyTorch, NumPy, Pandas, Scikit-learn, Hugging Face Transformers and Datasets
Research Methods	Benchmark and dataset construction, human evaluation, LLM-as-a-judge evaluation, statistical testing
Software Engineering	Next.js, React, React Native (Expo), Node.js, REST APIs, PostgreSQL, MongoDB
Systems & Tools	Linux, Git, Docker, AWS Lambda, LaTeX

Academic Honors & Certifications

- **2nd Runner-up** - "Bridging Engineering & Computer Science" International Hackathon, organized by ILM-AI, co-hosted by **St. George's, University of London**. Built a browser-based projectile motion simulator with optional air resistance (two-member team). GitHub | Live
- **Certifications:** AWS Cloud Practitioner · AWS Lambda · AWS DynamoDB · AWS API Gateway · Claude Code in Action (Anthropic) · Generative AI (Udacity & AWS) · ML Crash Course (Google).